

RESEARCH EXPERIENCE

My primary research focuses on instance-level perception in the open world. During my Ph.D., I concentrated on object detection within the context of autonomous driving, thoroughly investigating the two-dimensional challenges posed by out-of-distribution and domain shift. Specifically, I aimed to address practical issues such as encountering new environments (e.g., rainy or snowy conditions) and unknown objects.

- **Instance-level Perception and Recognition** 2020-Present
 - Object Detection; Instance Segmentation
- **Learning in the Open World with Out-of-distributed Samples and Distribution Shift** 2020-2024
 - Domain Adaptation; Open-set Recognition; Open-Vocabulary Detection
- **Learning with Latent Structure and Topology** 2020-2023
 - Causality; Graph-based Learning; Complex Networks
- **Generative Models** 2023-Present
 - Video Generation; Image Manipulation; Discriminative Tasks

In addition, I have participated as a co-author in research on the following topics. While I am not an expert in these areas, I have a foundational understanding that allows me to advance related progress quickly in the future.

- **3D Vision** 2023-2024
 - Gaussian Splatting; 3D Point Cloud Segmentation; Stereo Vision
- **Medical Imaging Analysis** 2021-Present
 - LLM Agent; MRI Brain Analysis; Tooth Analysis; Report Generation

EDUCATION

- **The Chinese University of Hong Kong (CUHK)** Feb. 2024 - Dec. 2024
 - Postdoc, Electronic Engineering
 - **Supervisor:** Prof. Yixuan Yuan
- **City University of Hong Kong (CityU)** Aug. 2020 - Dec. 2023
 - Ph.D., Electrical Engineering
 - **Supervisor:** Prof. Yixuan Yuan
 - **Early Graduation**
- **Tianjin University (TJU)** Sep. 2016 - Jun. 2020
 - B.S., Electrical Engineering
 - **GPA:** 3.83/4.0 (Top 2%)

WHAT MAKES ME DIFFERENT

Founder & Director of ScholaGO Education Technology Company Limited, Hong Kong [Link]

To gain a deeper understanding of technology, I **founded a startup company** with five shareholders to develop an innovative educational product, aiming to convert static knowledge into an interactive, multi-modal adventure immersively. Our company is supported by HKSTP, HK Tech 300, and Alibaba Cloud. My ultimate goal is to develop valuable technology and products to improve the national happiness index.

Research Highlights and Experience

- My works were selected as **Oral Presentations** at AAAI-22, MICCAI-22, CVPR-22, and ICCV-23. Additionally, I was honored as a **Best Paper Finalist** at CVPR-22 and had my work featured as a **Cover Paper** in ACS Photonics. I embarked on my research journey in 2020 and successfully defended my PhD in 2023 without prior master's experience.

Personal Quality and Character

- For myself, being a good person, exemplified by kindness, empathy, and open-mindedness, is much more important. I am happy that I have earned lots of appreciation from my co-authored students advised by me.

CONFERENCE PAPER

1. **CLIFF: Continual Latent Diffusion for Open-Vocabulary Object Detection**
Wuyang Li, Xinyu Liu, Jiayi Ma, Yixuan Yuan
European Conference on Computer Vision (**ECCV**), 2024
2. **Boosting Medical Image Analysis via Motion-Informed Generative Videos**
Wuyang Li, Xinyu Liu, Qiushi Yang, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
3. **Novel Scenes & Classes: Towards Adaptive Open-set Object Detection**
Wuyang Li, Xiaoqing Guo, Yixuan Yuan
IEEE International Conference on Computer Vision (**ICCV**), 2023, **Oral (1.8%)**

4. **Adjustment and Alignment for Unbiased Open Set Domain Adaptation**
Wuyang Li, Jie Liu, Bo Han, Yixuan Yuan
 IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023
5. **SIGMA: Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022, **Best Paper Finalist (0.4%)**
6. **SCAN: Cross Domain Object Detection with Semantic Conditioned Adaptation**
Wuyang Li, Xinyu Liu, Xiwen, Yao, Yixuan Yuan
 The Association for the Advance of Artificial Intelligence (**AAAI**), 2022, **Oral (4.5%)**
7. **Joint polyp detection and segmentation with heterogeneous endoscopic data**
Wuyang Li, Chen Yang, Jie Liu, Xiaoqing Guo, Yixuan Yuan
 International Symposium on Biomedical Imaging (**ISBI**) Workshop, 2021
8. **When 3D Partial Points Meets SAM: Tooth Point Cloud Segmentation with Sparse Labels**
 Yifan Liu, **Wuyang Li**, Cheng Wang, Hui Chen, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024, **Early Accept (14.0%)**
9. **DiffRect: Latent Diffusion Label Rectification for Semi-supervised Medical Image Segmentation**
 Xinyu Liu, **Wuyang Li**, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
10. **F2TNet: FMRI to T1w MRI Knowledge Transfer Network for Brain Multi-phenotype Prediction**
 Zhibin He, **Wuyang Li**, Yu Jiang, Zhihao Peng, Pengyu Wang, Xiang Li, Tianming Liu, Junwei Han, et al.
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
11. **LGS: A Light-weight 4D Gaussian Splatting for Efficient Surgical Scene Reconstruction**
 Hengyu Liu*, Yifan Liu*, Chenxin Li*, **Wuyang Li**, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
12. **Endora: Video Generation Models as Endoscopy Simulators**
 Chenxin Li, Hengyu Liu, Yifan Liu, Brandom Feng, **Wuyang Li**, Xinyu Liu, Zhen Chen, Jing Shao, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
13. **MRM: Masked Relation Modeling for Medical Image Pre-Training with Genetics**
 Qiushi Yang, **Wuyang Li**, Baopu, Li Yixuan Yuan
 IEEE International Conference on Computer Vision (**ICCV**), 2023
14. **H2GM: A Hierarchical Hypergraph Matching Framework for Brain Landmark Alignment**
 Zhibin He, **Wuyang Li**, Tuo Zhang, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2023
15. **Towards Robust Adaptive Object Detection under Noisy Annotations**
 Xinyu Liu, **Wuyang Li**, Qiushi Yang, Baopu Li, Yixuan Yuann
 IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022
16. **Intervention & Interaction Federated Abnormality Detection with Noisy Clients**
 Xinyu Liu, **Wuyang Li**, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2022, **Oral**

JOURNAL PAPER

1. **Bridge the Gap Between Visual and Linguistic Comprehension for Zero-shot Semantic Segmentation**
 Xiaoqing Guo*, **Wuyang Li***, Xinyu Liu, Yixuan Yuan (* Equal Contribution)
 International Journal of Computer Vision (**IJCV**), 2024, Under Revision
2. **SIGMA++: Improved Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2023
3. **SCAN++: Enhanced Semantic Conditioned Adaptation for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Transactions on Multimedia (**TMM**), 2022
4. **HTD: Heterogeneous Task Decoupling for Two-Stage Object Detection**
Wuyang Li, Zhen Chen, Baopu Li, Dingwen Zhang, Yixuan Yuan
 IEEE Transactions on Image Processing (**TIP**), 2021

5. **Stereo Vision Meta-Lens-Assisted Driving Vision**
Xiaoyuan Liu, Wuyang Li, Takeshi Yamaguchi, Zihan Geng, Takuo Tanaka, Din Ping Tsai, Mu Ku Chen
ACS Photonics, 2024, **Selected as Cover Paper**
6. **LLM-guided Decoupled Probabilistic Prompt for Continual Learning in Medical Image Diagnosis**
Yiwen Luo, Wuyang Li, Chen Cheng, Xiang Li, Tianming Liu, Yixuan Yuan
IEEE Transactions on Medical Imaging (**TMI**), 2024, Under Revision
7. **FM-APP: Foundation Model for Any Phenotype Prediction via fMRI to sMRI Knowledge Transfer**
Zhibin He, Wuyang Li, Yifan Liu, Xinyu Liu, Junwei Han, Tuo Zhang, Yixuan Yuan
IEEE Transactions on Medical Imaging (**TMI**), 2024, Under Review
8. **Medical Federated Learning with Joint Graph Purification for Noisy Label Learning**
Zhen Chen, Wuyang Li, Xiaohan Xing, Yixuan Yuan
Medical Image Analysis (**MIA**), 2023
9. **Decoupled Unbiased Teacher for Source-free Domain Adaptive Medical Object Detection**
Xinyu Liu, Wuyang Li, Yixuan Yuan
IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**), 2023
10. **GRAB-Net: Graph-based Boundary-aware Network for Medical Point Cloud Segmentation**
Yifan Liu, Wuyang Li, Jie Liu, Hui Chen, Yixuan Yuan
IEEE Transactions on Medical Imaging (**TMI**), 2023

PREPRINT

1. **U-KAN Makes Strong Backbone for Medical Image Segmentation and Generation**
Chenxin Li*, Xinyu Liu*, Wuyang Li*, Cheng Wang*, Hengyu Liu, Yixuan Yuan (* Equal Contribution)
ArXiv preprint arXiv:2406.02918, 2024

RESEARCH EXPERIENCE

- **The Chinese University of Hong Kong** Feb. 2024 - Dec. 2024
Image/video generative models.
Supervisor: Prof. Yixuan Yuan
- **City University of Hong Kong** Aug. 2020 - Dec. 2023
Develop domain/noise-robust generic object detectors.
Supervisor: Prof. Yixuan Yuan
- **NUS (Suzhou) Research Institute (NUSRI)** Sep. 2019 - Jun. 2020
Develop a web-page-based Augmentation Reality (AR) framework.
Supervisor: Prof. Zhiying, Zhou

SELECTED AWARDS

- **Outstanding Academic Performance Award (OAPA), CityU** 2022-2023
Outstanding research achievements in the EE department.
- **Research Tuition Scholarship (RTS), CityU** 2022-2023
Outstanding research achievements in the EE department.
- **Ranked First in Polyp Detection Challenge** May 2021
3rd International Endoscopy Computer Vision Challenge and Workshop (EndoCV2021)
- **National Scholarship** 2017-2018
The Highest Honor for a Student in China (Top 2%)
- **Silver Award in Tianjin Advanced Mathematical Competition** Jun 2017
As a Selected Participant Representing Tianjin University (Top 5%)
- **First-class Student Awards** 2017-2019
Outstanding Students Selected in Tianjin University (Top 10%)

SERVICES

- **T-PAMI; IJCV; TIP; TNNLS; T-IV; T-ASE; PR** -
Journal Reviewer
- **NeurIPS-24; CVPR-23/24; ICCV-23; AAAI-23/24** -
Conference Reviewer